

backend/api/agent.py Documentation (Roman Urdu)

Yeh file Project ke Next.js Frontend (ya kisi bb client) aur **Agent Framework** ke darmyan bridge (pul) ka kam krti hy. Is file me FastAPI k REST Routes likhay gay hain.

File Ka Maqsad (Purpose)

Is script se user UI me dekhta hai k konsey Tools available hain (Tool Registry) ar ReAct Agent ko API call lga kr Sawal bhej skta ahi (/api/agent/run).

Code Blocks aur API Endpoints ki Tafseel

1. Tool Endpoints

- **GET /api/agent/tools**: Yeh route Frontend ko saray built-in tools (jaisay math_calculator, web_search, file_reader) ki details JSON array mein return krta ha. Frontend isay Apni Agent View me dikhata hai.
- **POST /api/agent/execute**: Agar user Agent ki bajaye manually sirf tool test karna chahat ha tou is route p call marta hai.

2. Agent Run Endpoints

- **POST /api/agent/run**: Normal response dta ha waiting k baad jb Agent pura sooch k ans nkal chuka h.
- **POST /api/agent/run/stream (Server-Sent Events)**: Yeh real-time live connection ha. Jab AI "Sochta ha" **Thinking...**, phr **Web Search Action...**, then **Observation...** yeh sab steps UI ko animation/stream k zariye foran show krne m mdd data ha! Bad mein chizein db m (Database) me save kr di jati hain.

3. Memory Endpoints

- **GET / POST / DELETE /api/agent/memory**: AI ne jo jo chizaien note kar kay apni memory m rakhi hyn (ta k mustaqbik m yad rahen) user unhe iss endpoint se dekh sakta, delete sikh skta ya naya add kr skyta.